
INTRO

High-Resolution Image Synthesis with Latent Diffusion Models

Stable Diffusion

High-Resolution Image Synthesis with Latent Diffusion Models

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INTRO

High-Resolution Image Synthesis with Latent Diffusion Models

Text-to-Image Synthesis on LAION. 1.45B Model.

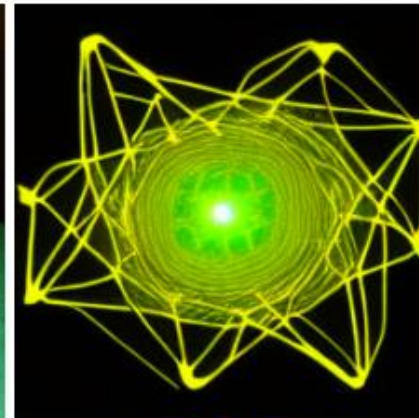
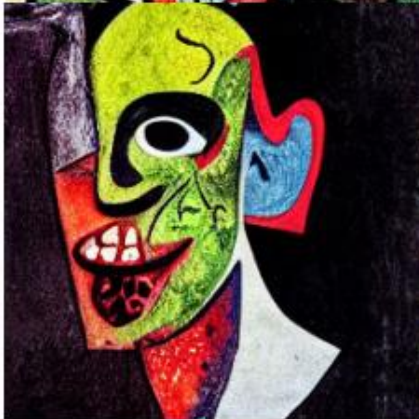
*'A street sign that reads
"Latent Diffusion" '*

*'A zombie in the
style of Picasso'*

*'An image of an animal
half mouse half octopus'*

*'An illustration of a slightly
conscious neural network'*

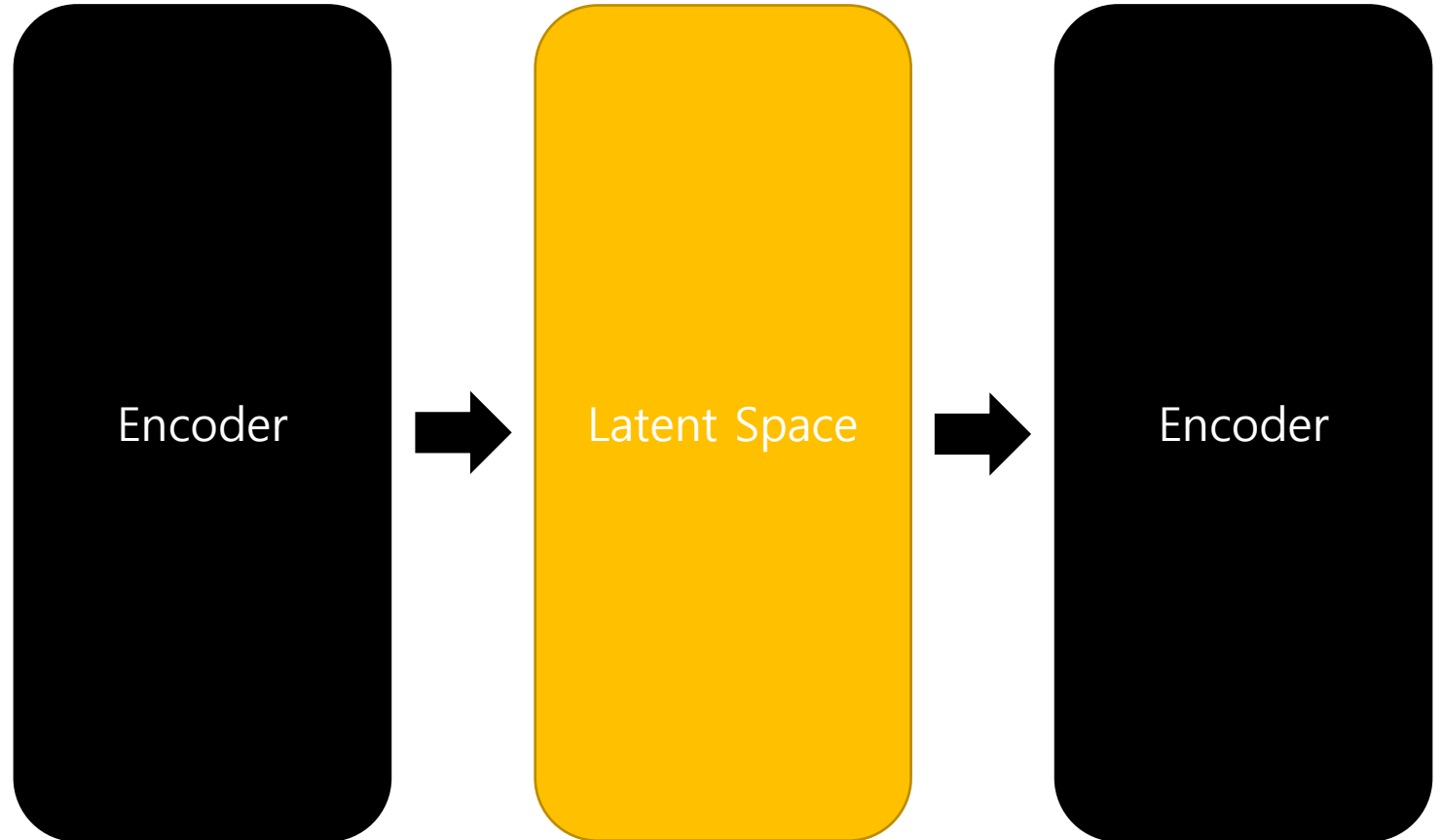
*'A painting of a
squirrel eating a burger'*



PREREQUISITES

Auto Encoder & Variational Auto Encoder

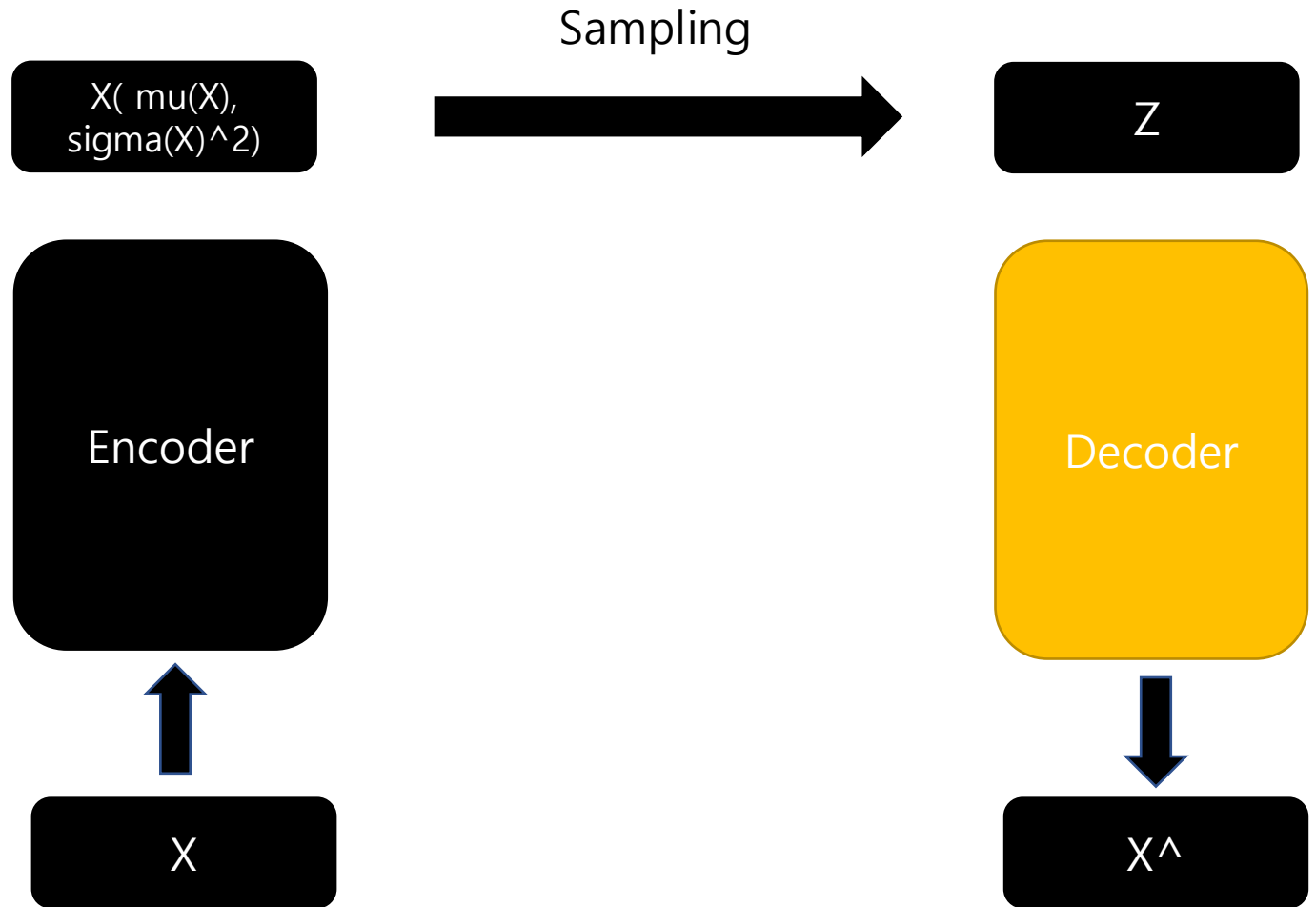
- 이미지와 같은 고차원 데이터를 Encoding을 통해서 저차원 Hidden 공간으로 표현 ..
- 이러한 의미 있는 정보를 담은 차원 공간을 Latent Space, 정보를 담은 변수를 Latent Variable 이라고 부른다.
- Encoder 를 이용해서 input 을 Latent Space 로 맵핑시키는 과정은 PCA와 같은 성분 분해의 개념과 유사
- Decoder 에서는 이러한 주요한 요소들을 다시 복원 시키는 역할



PREREQUISITES

Auto Encoder & Variational Auto Encoder

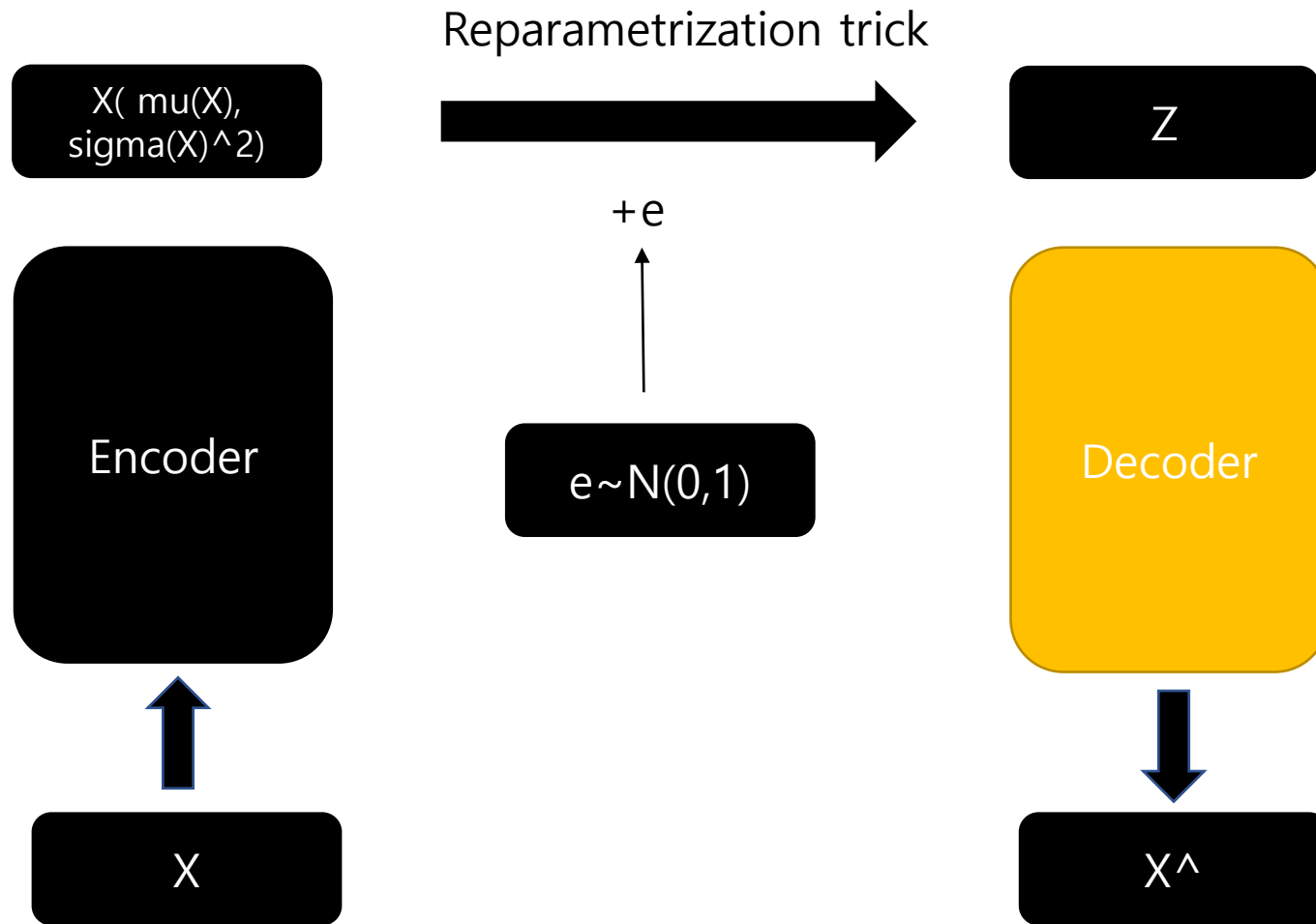
- X 데이터가 너무 고차원 이므로, 이를 다시 작은 분포로 변환, 예를 들어서 정규분포로 변환할 때는 평균과 분산을 이용해서 분포를 표현 가능



PREREQUISITES

Auto Encoder & Variational Auto Encoder

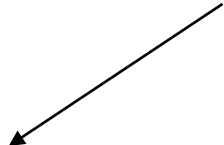
- Reparametrization trick 을 활용해서, 가우시안 분포에서 바로 분포를 추정하는 것이 아니라, noise 값을 더해서 미분가능한 함수로 바꿔준 후에 학습



PREREQUISITES

Auto Encoder & Variational Auto Encoder

- **VAE ELBO**
: Evidence lower Bound

$$\begin{aligned}\log p_{\theta}(\mathbf{x}) &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}) d\mathbf{z} \\ &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})} d\mathbf{z} \\ &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})} \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{q_{\phi}(\mathbf{z}|\mathbf{x})} d\mathbf{z} \\ &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - KL(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) + KL(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z}|\mathbf{x}))\end{aligned}$$


- Bayes Rule 조건부 확률 식을 이용하여, 샘플 x_i 가 발생할 확률을 치환
- 모든 샘플 x_i 에 대한 샘플 z 가 생성될 확률 밀도함수의 적분

- **Kullback-Leiber divergence**

$$D_{KL}(P||Q) = \sum_{x \in \chi} P(x) \log_b \left(\frac{P(x)}{Q(x)} \right)$$

PREREQUISITES

Auto Encoder & Variational Auto Encoder

- VAE ELBO
: Evidence lower Bound

$$\log p_{\theta}(\mathbf{x}) \geq \underbrace{\int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z}}_{\text{Reconstruction error}} - \underbrace{KL(q_{\phi}(\mathbf{z}|\mathbf{x}) || p(\mathbf{z}))}_{\text{Regularization error}} \leftarrow \text{Evidence lower bound (ELBO)}$$

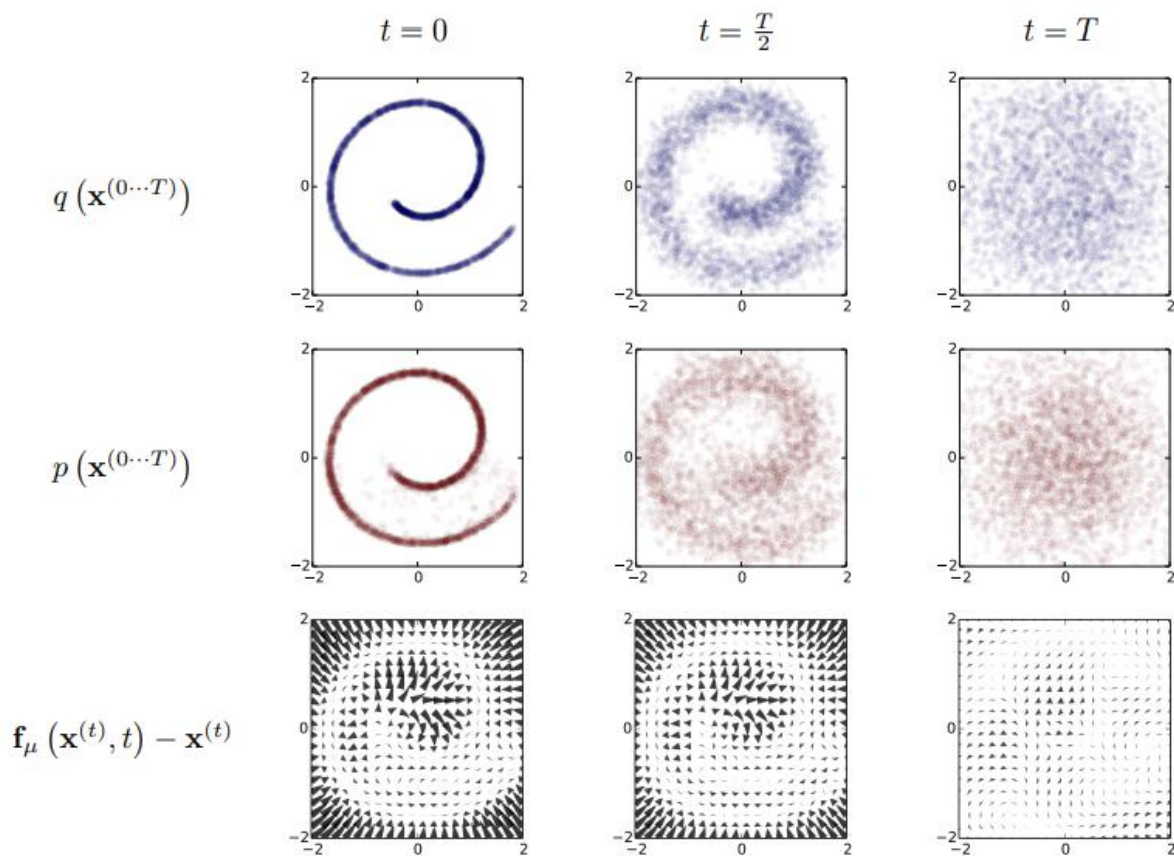
- Z로부터 x가 나타날 확률을 최대화 하는 것으로 MLE(Maximum Likelihood Estimation) dl 되며 이는 Cross-Entropy로 표현됨

PREREQUISITES

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

▪ Forward Diffusion Process

- 실제 데이터 분포 $q(\mathbf{x}(0))$ 를 forward process 를 통해서 $\pi(\mathbf{y})$ 로 바꾸게 된다
- 이 과정을 Markov Diffusion Kernel 이라고 한다



$$\pi(\mathbf{y}) = \int d\mathbf{y}' T_\pi(\mathbf{y} | \mathbf{y}'; \beta) \pi(\mathbf{y}')$$
$$q(\mathbf{x}^{(t)} | \mathbf{x}^{(t-1)}) = T_\pi(\mathbf{x}^{(t)} | \mathbf{x}^{(t-1)}; \beta_t)$$
$$q(\mathbf{x}^{(0\cdots T)}) = q(\mathbf{x}^{(0)}) \prod_{t=1}^T q(\mathbf{x}^{(t)} | \mathbf{x}^{(t-1)})$$

Bt 를 조절함으로써 변환단계의 길이를
조절 가능

PREREQUISITES

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

- **Forward Diffusion Process => Reverse Trajectory**

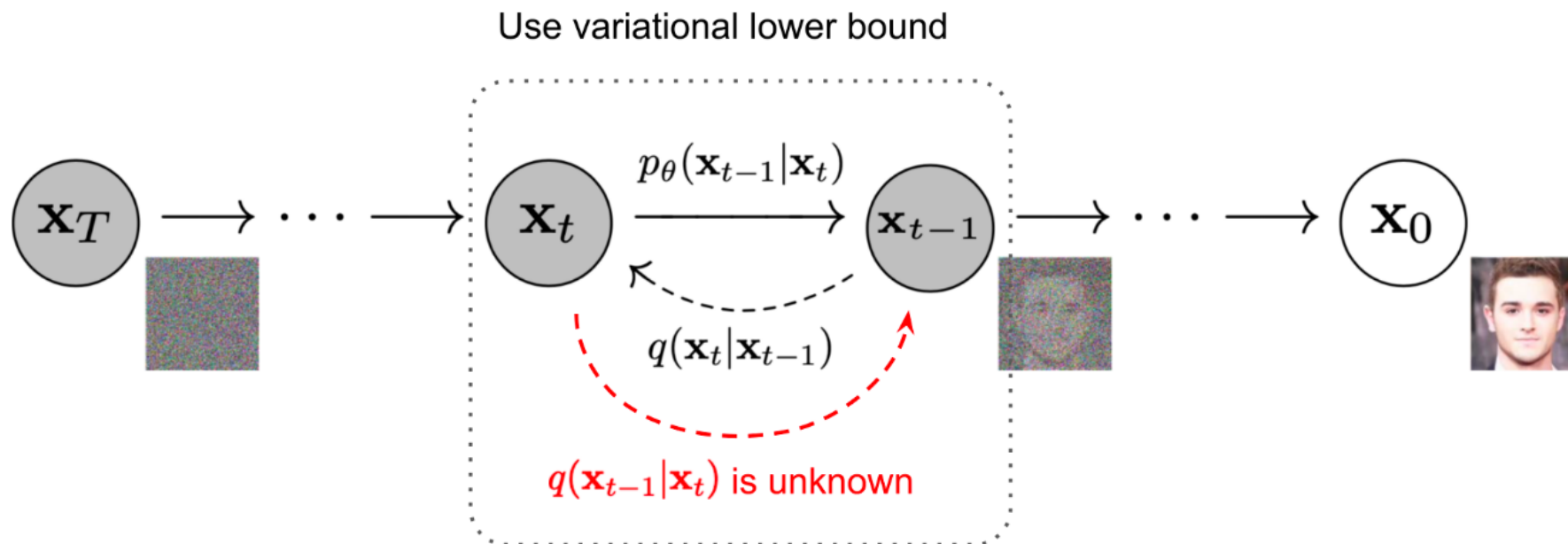
$$p\left(\mathbf{x}^{(T)}\right)=\pi\left(\mathbf{x}^{(T)}\right)$$

$$p\left(\mathbf{x}^{(0 \cdots T)}\right)=p\left(\mathbf{x}^{(T)}\right) \prod_{t=1}^T p\left(\mathbf{x}^{(t-1)} \mid \mathbf{x}^{(t)}\right)$$

- Binomial 이나 Gaussian 의 경우는 , Step Size B가 매우 작다면 reverse diffusion process 가 forward 와 같다.
- [Feller's Contributions to the One-Dimensional Diffusion Theory and Beyond]
- Reverse Process 를 유도하는 수식을 참고하면 Bt 를 결정하는 것이 성능에 매우 중요한 역할

PREREQUISITES

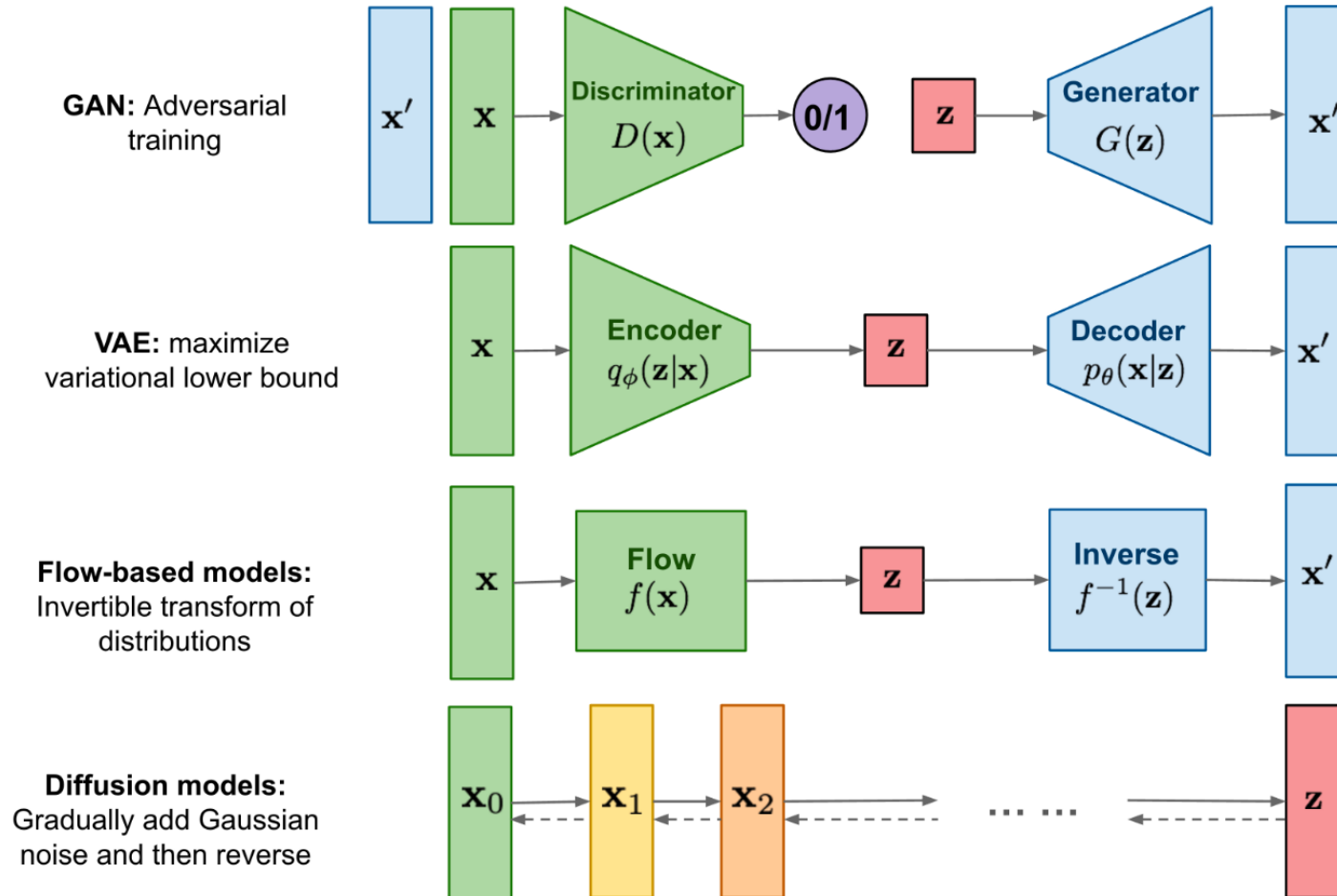
DDPM



$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}), \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

PREREQUISITES

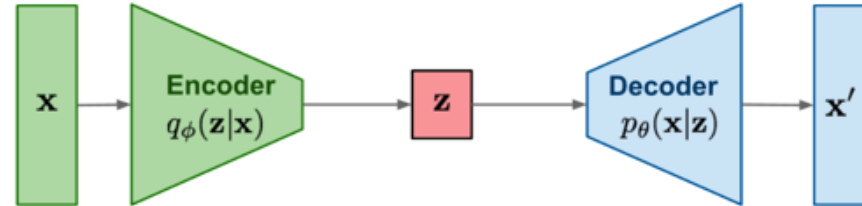
GAN & VAE & DM



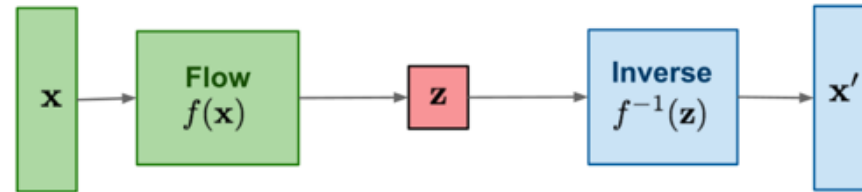
PREREQUISITES

High-Resolution Image Synthesis with Latent Diffusion Models

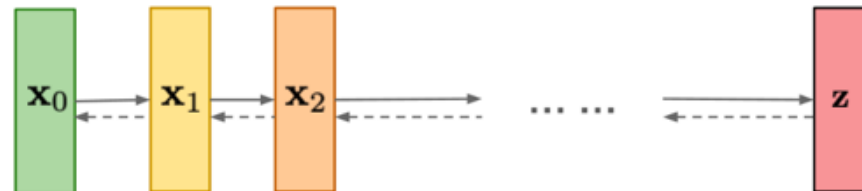
VAE: maximize
variational lower bound



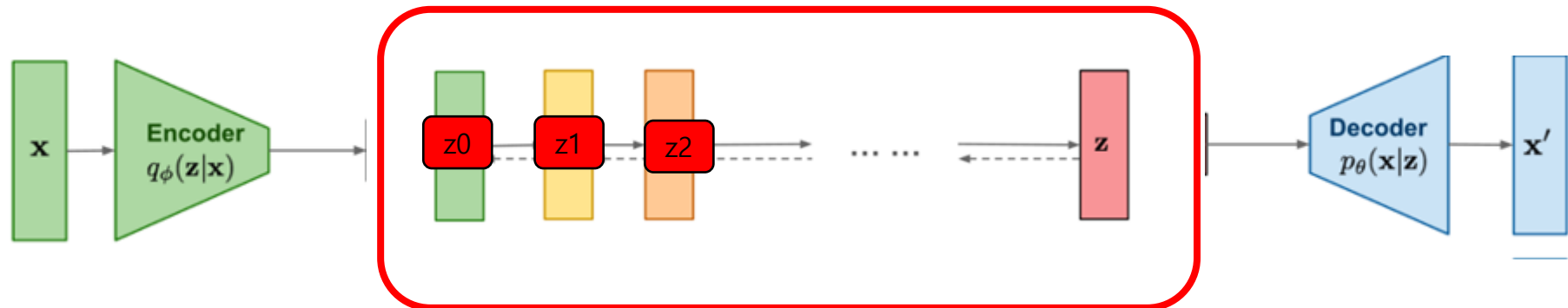
Flow-based models:
Invertible transform of
distributions



Diffusion models:
Gradually add Gaussian
noise and then reverse



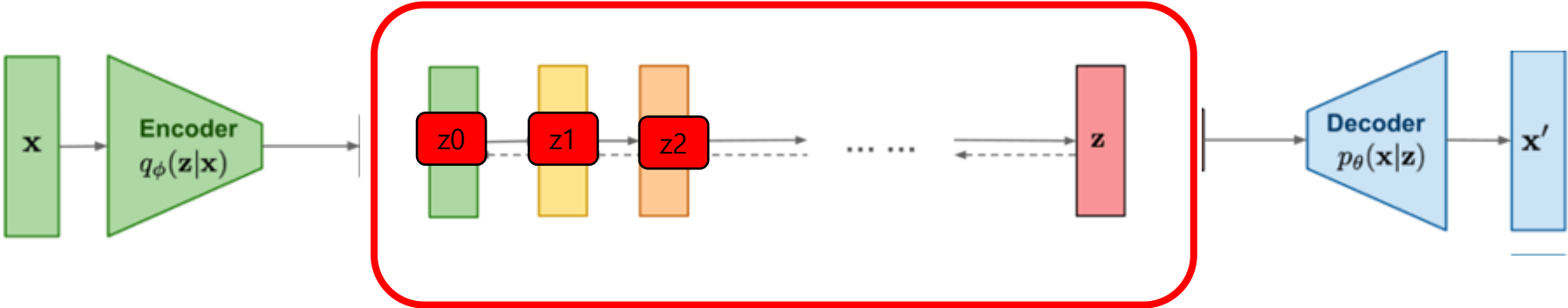
Stable Diffusion



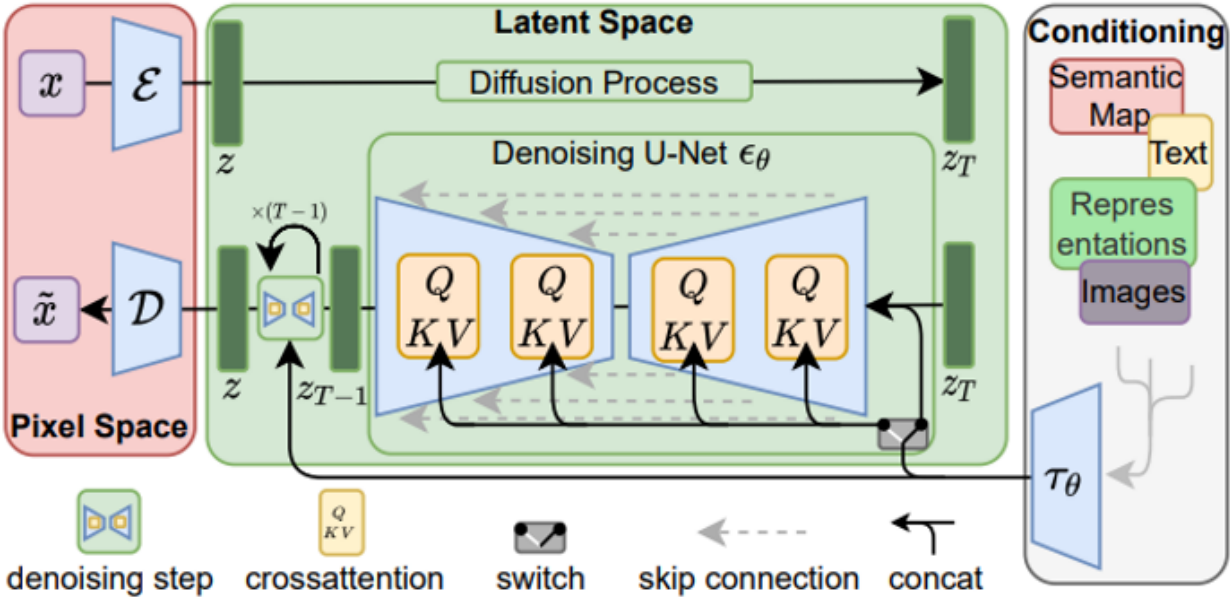
ARCHITECTURE

High-Resolution Image Synthesis with Latent Diffusion Models

Stable Diffusion



Architecture



KEY POINT

High-Resolution Image Synthesis with Latent Diffusion Models

- Denoising 과정에서 AutoEncoder 를 사용
- Pixel 공간이 아닌, Latent Space 에서 Denoising 을 진행하면서 수백개의 GPU 가 필요한 컴퓨팅 코스트를 줄임
- 모델의 복잡성 감소, 세부적인 표현능력 상승
- 아키텍처 상에 Cross-Attention을 사용함으로써, 다른 도메인(Text, Audio ..) 을 함께 모델상에서 사용가능

1. INTRODUCTION

Democratizing High-Resolution Image Synthesis

- Diffusion Model 들은 컴퓨팅 코스트를 많이 필요로함. 학습과 평가 과정에 있어서 반복적인 함수의 연산을 고차원의 공간(RGB images) 에서 해야하기 때문에.
- 예시로, 가장 많은 연산을 필요로 하는 DM 의 경우에 150~1000대의 V100 을 수일간 학습하기도 함
=> Prafulla Dhariwal and Alex Nichol. Diffusion models beat gans on image synthesis. CoRR, abs/2105.05233, 2021. 1,

위의 케이스의 경우 50k 샘플들의 추론에만 A100 한 대로, 5일이 걸림

- 결론적으로 학습과 추론 모두 너무 많은 컴퓨팅 코스트를 필요로함
- 따라서 Diffusion 의 Model 의 성능을 살리면서 연산량을 줄이는 것이 목표였음

1. INTRODUCTION

High-Resolution Image Synthesis with Latent Diffusion Models

Importantly, the encoder downsamples the image by a factor $f = H/h = W/w$



Figure 1. Boosting the upper bound on achievable quality with less aggressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512^2 px. We denote the spatial downsampling factor by f . Reconstruction FIDs [29] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

1. INTRODUCTION

High-Resolution Image Synthesis with Latent Diffusion Models

FID (Frechet Inception Distance)

- Inception-V3 모델을 feature extractor 로 사용하여, 실제 이미지와 생성된 이미지 사이에 Activation Map 을 추출
- Activation Map 을 통해서 multivariate gaussian distribution 을 구해서 두 분포 사이의 Wassertein-2 distance 를 측정

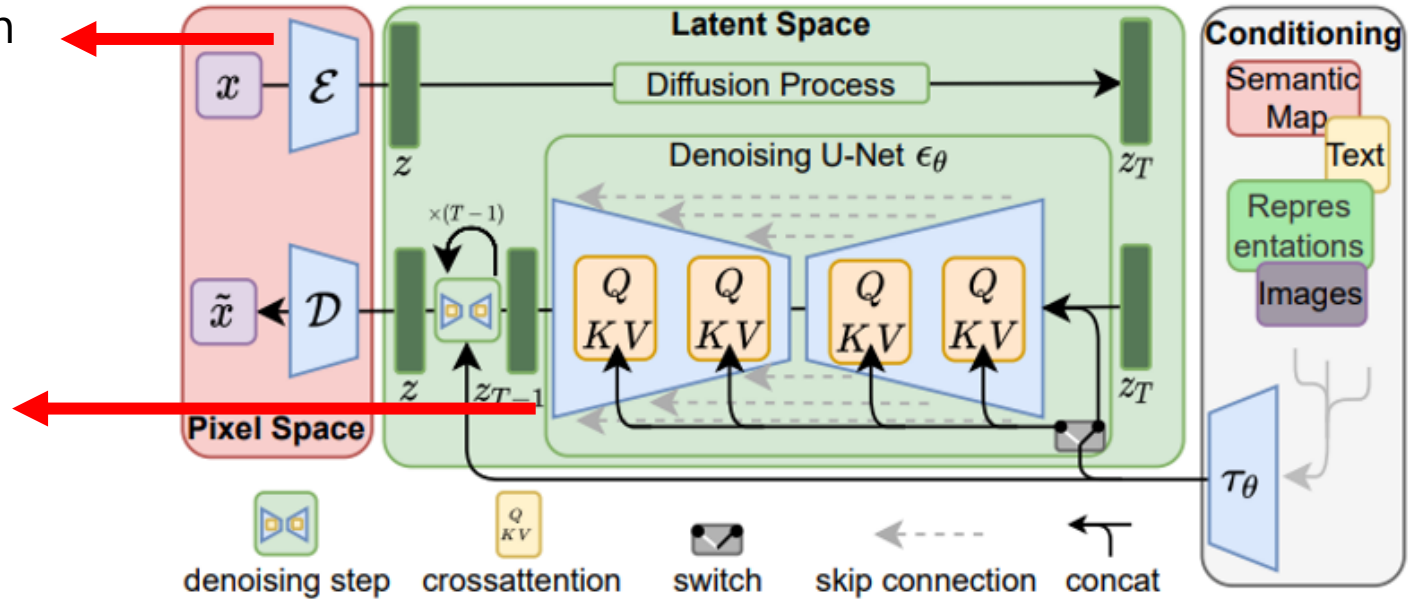
$$FID = d^2 = \|\mu_1 - \mu_2\|_2^2 - Tr(\Sigma_1 + \Sigma_2 - 2\Sigma_1\Sigma_2)$$

μ, Σ 는 각각 평균과 Covariance Matrix 를 의미

1. INTRODUCTION

High-Resolution Image Synthesis with Latent Diffusion Models

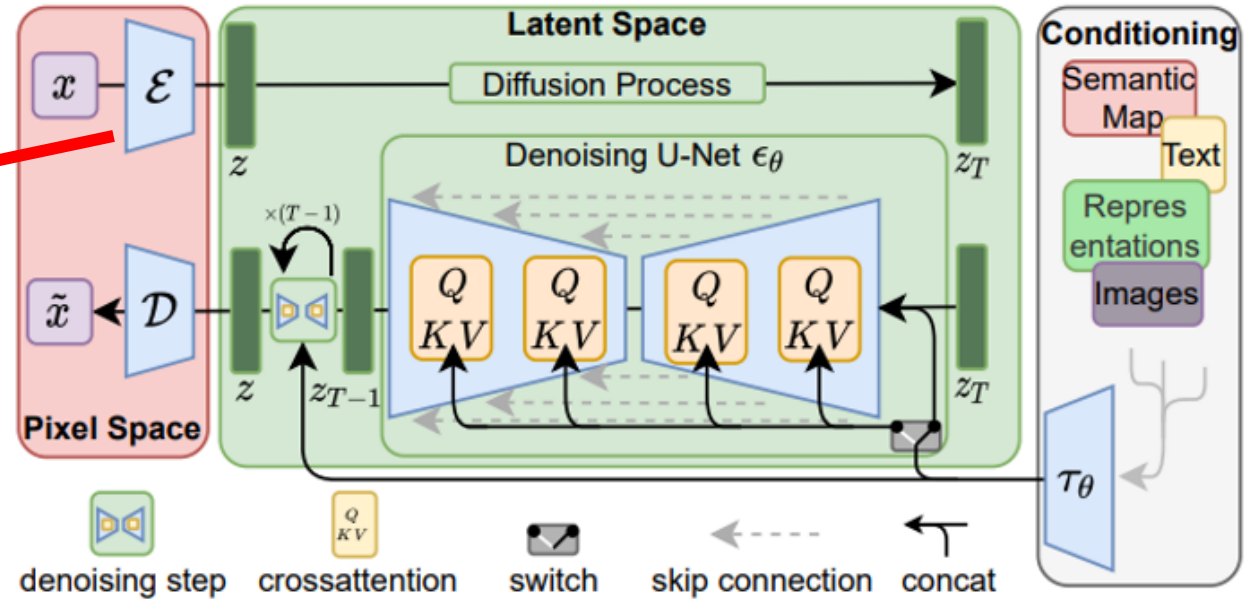
- First, we train an **autoencoder** which provides a lower-dimensional (and thereby efficient)
- we do not need to rely on excessive spatial compression, as **we train DMs in the learned latent space**, which exhibits better scaling properties with respect to the spatial dimensionality
- We dub the resulting model class **Latent Diffusion Models (LDMs)**.



1. INTRODUCTION

High-Resolution Image Synthesis with Latent Diffusion Models

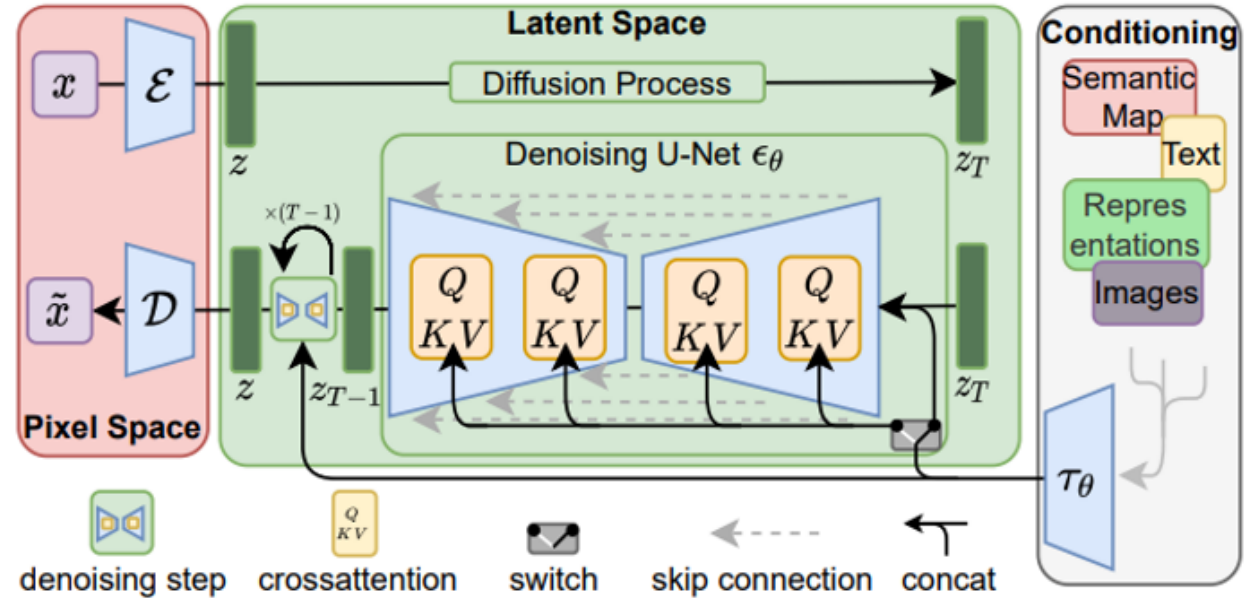
- A notable advantage of this approach is that we need to train the universal autoencoding stage only once and can therefore **reuse it for multiple DM trainings** or to explore possibly completely different tasks



1. INTRODUCTION

High-Resolution Image Synthesis with Latent Diffusion Models

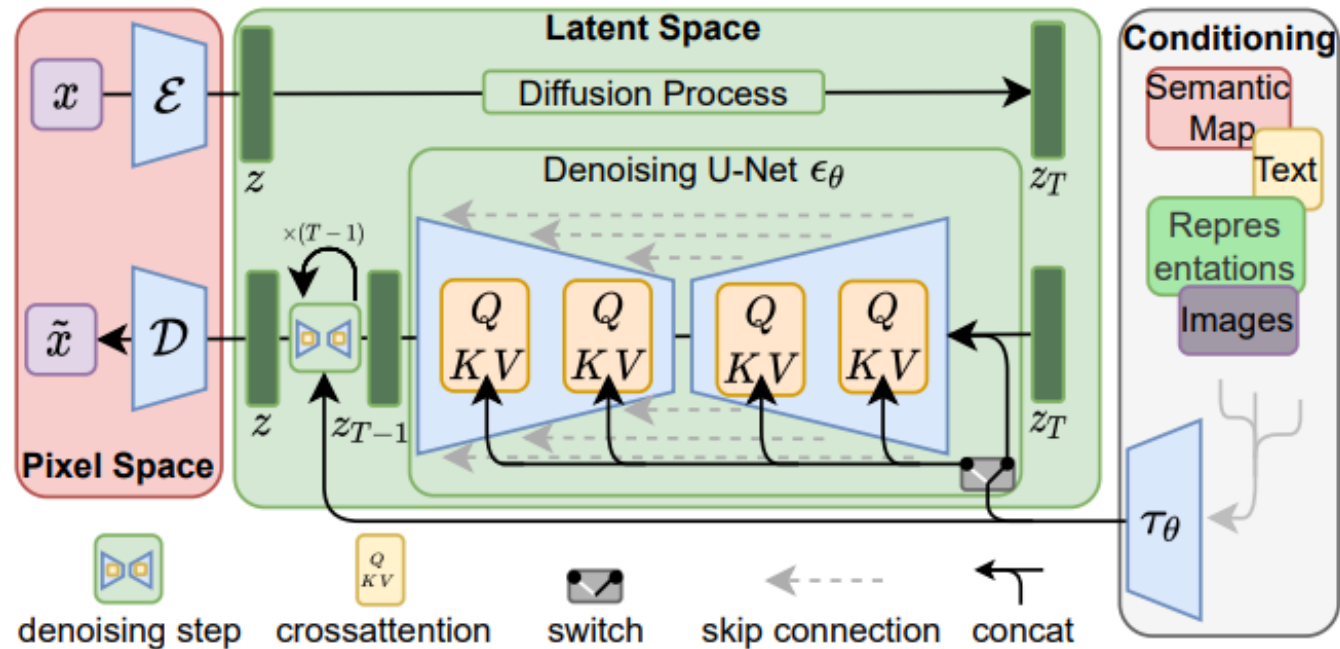
- We achieve competitive performance on multiple tasks (**unconditional image synthesis, inpainting, stochastic super-resolution**) and datasets while significantly lowering **computational costs**.
- Moreover, we design a general-purpose conditioning mechanism based on **cross-attention**, enabling multi-modal training. We use it to train class-conditional, text-to-image and layout-to-image models



3. Method

3.2. Latent Diffusion Models

Generative Modeling of Latent Representations



Latent Space (Low dimensional)

Likelihood-based generative models 로써, High Dimensional 한 Pixel Space 보다 Low Dimension 인 Latent Space 에서의 연산이 훨씬 유리하다.

- a) 중요하고, 시멘틱한 포인트 들에 더 집중할 수 있다.
- b) low dimension 에서 계산을 진행함으로써, 컴퓨팅 코스트를 많이 줄일 수 있다

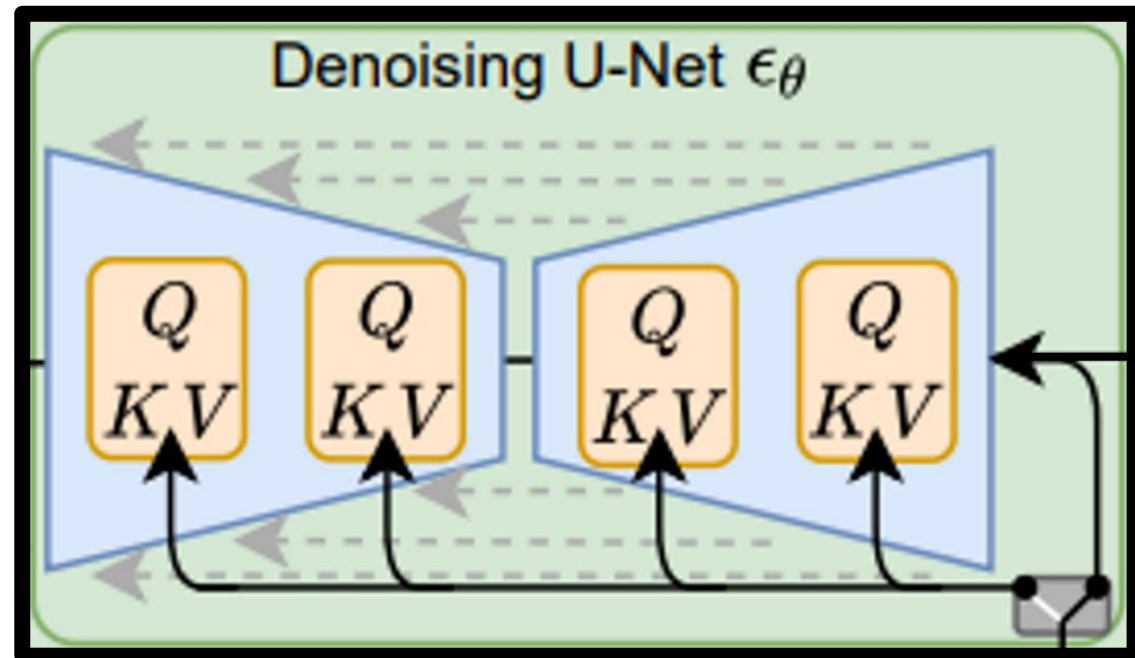
Underlying Unet

Inductive bias 를 활용할 수 있음
Time-conditional Unet

Figure 3. We condition LDMs either via concatenation or by a more general cross-attention mechanism. See Sec. 3.3

3. Method

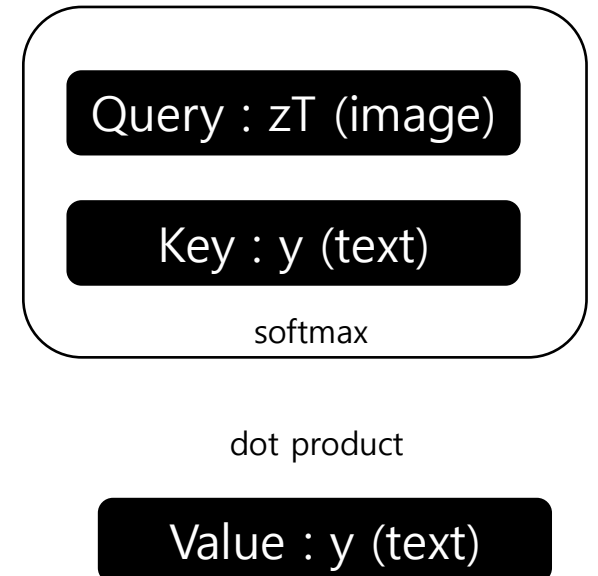
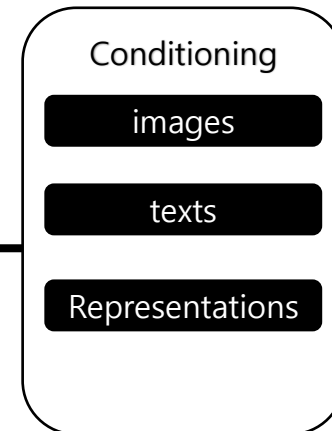
3.3. Conditioning Mechanisms



▪ Cross-Attention Mechanism

$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) \cdot V$, with

$$Q = W_Q^{(i)} \cdot \varphi_i(z_t), \quad K = W_K^{(i)} \cdot \tau_\theta(y), \quad V = W_V^{(i)} \cdot \tau_\theta(y).$$



3. Method

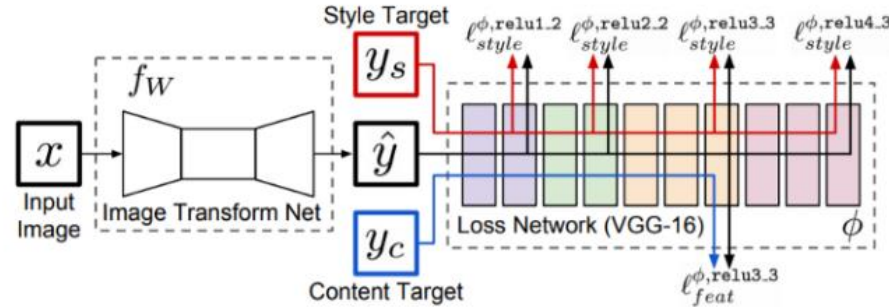
3.1. Perceptual Image Compression

Perceptual loss



A patch-based
adversarial objective

- [medium] Perceptual Loss — Well it sounds interesting after Neural Style Transfer



$$\ell_{feat}^{\phi,j}(\hat{y}, y) = \frac{1}{C_j H_j W_j} \|\phi_j(\hat{y}) - \phi_j(y)\|_2^2$$

- [33] Image-to-image translation with conditional adversarial networks
- 이미지 전체를 이용해서 판별하지 않고, 이미지 내의 패치를 특정 단위로 잘라서 True/False 를 판별
- 지역적인 사실성을 살릴 수 있음 (Enforcing local realism)
- L1, L2 Loss 처럼 픽셀 단위의 Loss 에 집중했을 때, 나타날 수 있는 흐림 현상(Blurriness) 을 완화

4. Experiments

4.2. Image Generation with Latent Diffusion

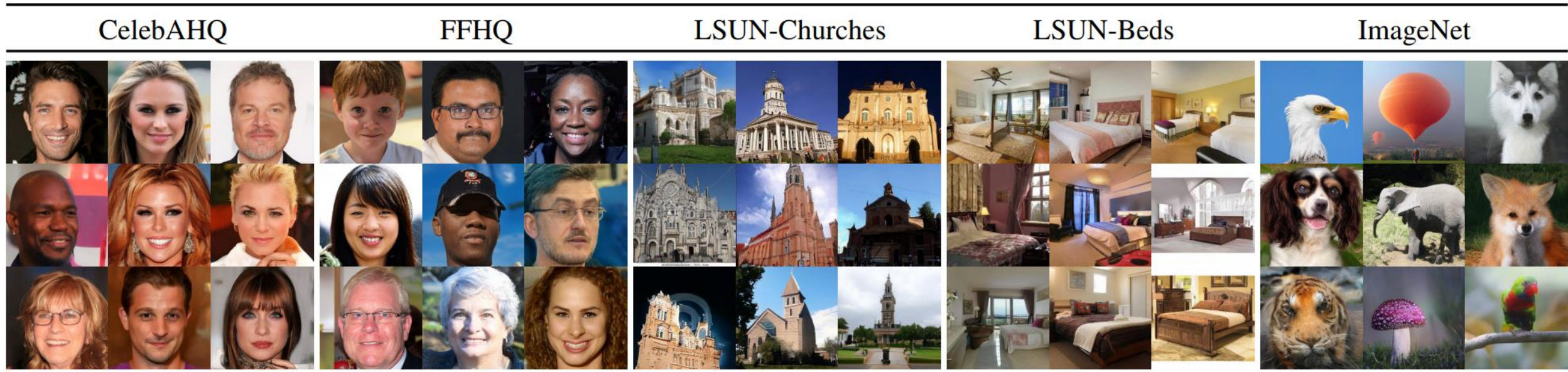


Figure 4. Samples from *LDMs* trained on CelebAHQ [39], FFHQ [41], LSUN-Churches [102], LSUN-Bedrooms [102] and class-conditional ImageNet [12], each with a resolution of 256×256 . Best viewed when zoomed in. For more samples *cf.* the supplement.

4. Experiments

4.2. Image Generation with Latent Diffusion

CelebA-HQ 256×256				FFHQ 256×256			
Method	FID ↓	Prec. ↑	Recall ↑	Method	FID ↓	Prec. ↑	Recall ↑
DC-VAE [63]	15.8	-	-	ImageBART [21]	9.57	-	-
VQGAN+T. [23] (k=400)	10.2	-	-	U-Net GAN (+aug) [77]	10.9 (7.6)	-	-
PGGAN [39]	8.0	-	-	UDM [43]	5.54	-	-
LSGM [93]	7.22	-	-	StyleGAN [41]	<u>4.16</u>	<u>0.71</u>	<u>0.46</u>
UDM [43]	<u>7.16</u>	-	-	ProjectedGAN [76]	3.08	0.65	<u>0.46</u>
<i>LDM-4</i> (ours, 500-s [†])	5.11	0.72	0.49	<i>LDM-4</i> (ours, 200-s)	4.98	0.73	0.50

LSUN-Churches 256×256				LSUN-Bedrooms 256×256			
Method	FID ↓	Prec. ↑	Recall ↑	Method	FID ↓	Prec. ↑	Recall ↑
DDPM [30]	7.89	-	-	ImageBART [21]	5.51	-	-
ImageBART [21]	7.32	-	-	DDPM [30]	4.9	-	-
PGGAN [39]	6.42	-	-	UDM [43]	4.57	-	-
StyleGAN [41]	4.21	-	-	StyleGAN [41]	2.35	0.59	<u>0.48</u>
StyleGAN2 [42]	<u>3.86</u>	-	-	ADM [15]	<u>1.90</u>	0.66	0.51
ProjectedGAN [76]	1.59	<u>0.61</u>	<u>0.44</u>	ProjectedGAN [76]	1.52	<u>0.61</u>	0.34
<i>LDM-8*</i> (ours, 200-s)	4.02	0.64	0.52	<i>LDM-4</i> (ours, 200-s)	2.95	0.66	<u>0.48</u>

Table 1. Evaluation metrics for unconditional image synthesis.

4. Experiments

4.3. Conditional Latent Diffusion

Text-Conditional Image Synthesis				
Method	FID ↓	IS↑	N_{params}	
CogView [†] [17]	27.10	18.20	4B	self-ranking, rejection rate 0.017
LAFITE [†] [109]	26.94	<u>26.02</u>	75M	
GLIDE* [59]	<u>12.24</u>	-	6B	277 DDIM steps, c.f.g. [32] $s = 3$
Make-A-Scene* [26]	11.84	-	4B	c.f.g for AR models [98] $s = 5$
<i>LDM-KL-8</i>	23.31	20.03 ± 0.33	1.45B	250 DDIM steps
<i>LDM-KL-8-G*</i>	12.63	30.29 ± 0.42	1.45B	250 DDIM steps, c.f.g. [32] $s = 1.5$

Table 2. Evaluation of text-conditional image synthesis on the 256×256 -sized MS-COCO [51] dataset: with 250 DDIM [84] steps our model is on par with the most recent diffusion [59] and autoregressive [26] methods despite using significantly less pa-
rameters. [†]/*:Numbers from [109]/[26]

layout-to-image synthesis on the COCO dataset

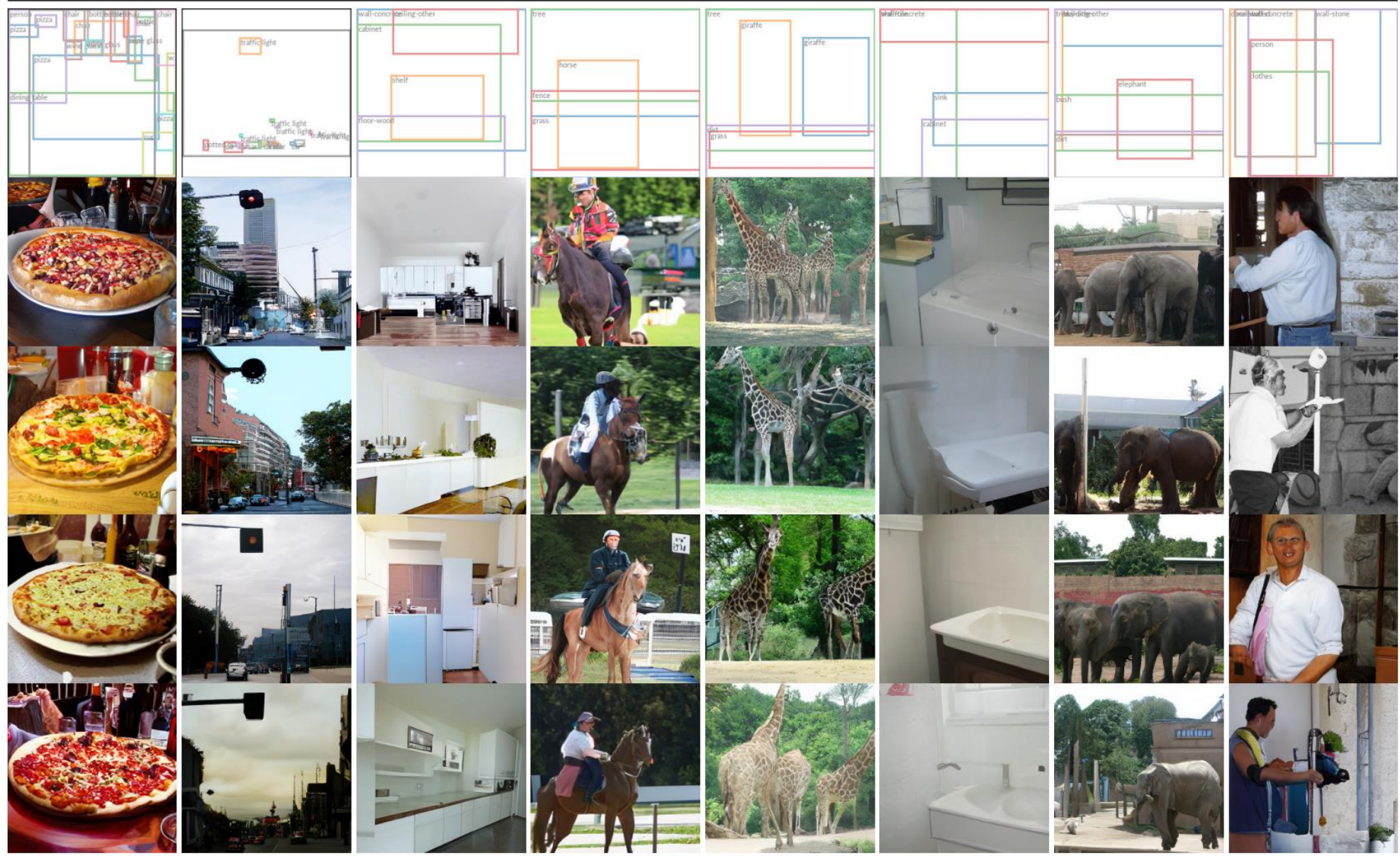
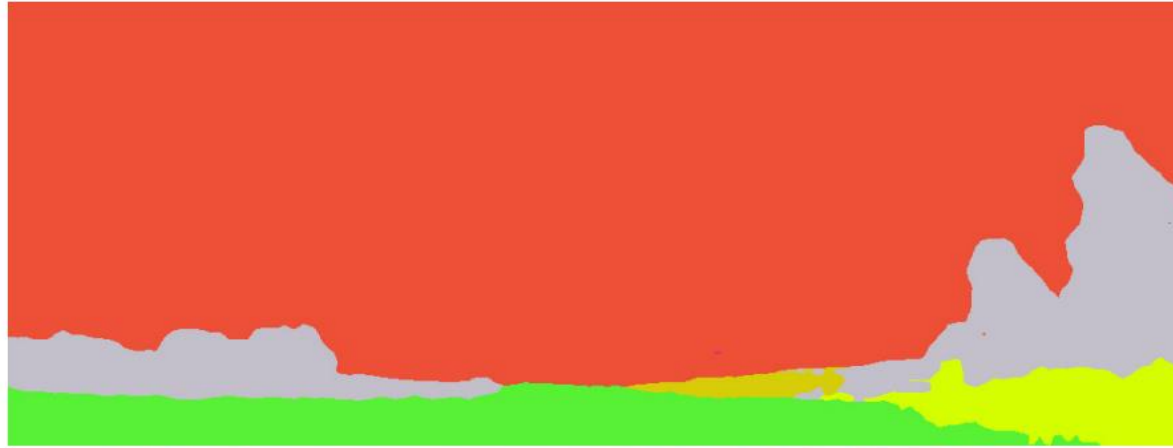


Figure 16. More samples from our best model for layout-to-image synthesis, *LDM-4*, which was trained on the OpenImages dataset and finetuned on the COCO dataset. Samples generated with 100 DDIM steps and $\eta = 0$. Layouts are from the COCO validation set.

Image inpainting





End

High-Resolution Image Synthesis with Latent Diffusion Models

Thank you